



Background

The Challenge

Individuals with ADHD often struggle with meal planning and dietary management due to executive function challenges, decision fatigue, and difficulty maintaining consistent routines. Traditional meal planning apps lack the specialized support needed for this community.

Our Solution

Mind Meal is a mobile-first application designed specifically for ADHD users, providing personalized recipe suggestions based on available inventory, health needs, and dietary goals. The app reduces cognitive load through intuitive design, smart notifications, and conversational recipe discovery.

Key Innovation

By combining inventory awareness with AI-powered recipe suggestions and ADHD-friendly interface design, Mind Meal transforms meal planning from an overwhelming task into a manageable daily routine.

Key Requirements

User-Centric Design

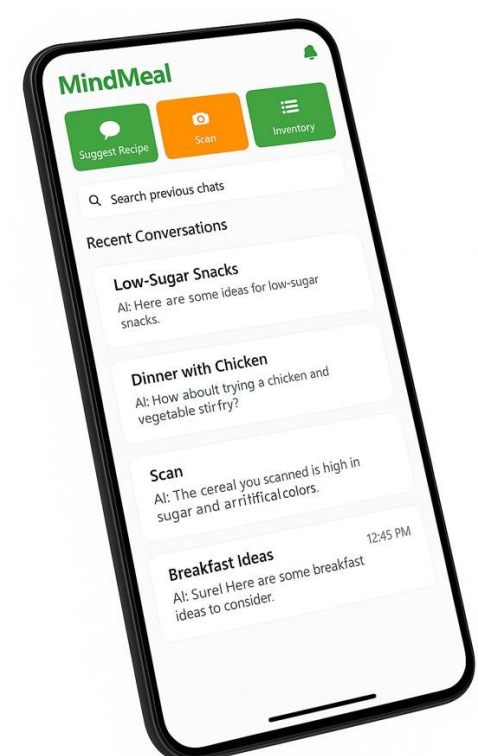
Simple, distraction-free interface optimized for ADHD users
Minimal clicks required for common tasks
Clear visual hierarchy and progress indicators

Personalization Engine

Recipe suggestions filtered by allergies, medical conditions, and dietary goals
Inventory-based meal recommendations
Contextual notifications for expiring items and meal reprints

Core Functionality

Streamlined onboarding to capture health and dietary preferences
Real-time inventory management (add/update/delete items)
Conversational recipe discovery through chat interface
Smart notification system for reminders and alerts



Architectural Design

Three-Layer Architecture

The system follows a modular design separating concerns across three distinct layers:

UI Layer - MindMeal components including:

Login/Registration with social auth options
Multi-step onboarding with progress tracking
Dashboard-style home page with quick navigation
Visual inventory management with floating action buttons

Chat-based recipe maker interface

Grouped notification center

Camera module (planned) for barcode scanning

Application Logic Layer - Core business logic:

Authentication and session management

Onboarding data validation and storage

Recipe engine with allergen filtering and dietary constraints

Notification scheduling and categorization

Inventory rules and expiration tracking

Personal information management across modules

Data Layer - Persistent storage and external services:

Local storage for offline-first functionality

API client for LLM and external service communication

Push notification service integration

Future integrations: barcode scanning, nutrition databases, cloud sync

Data Flow Architecture

The modular design enables clear information flow:

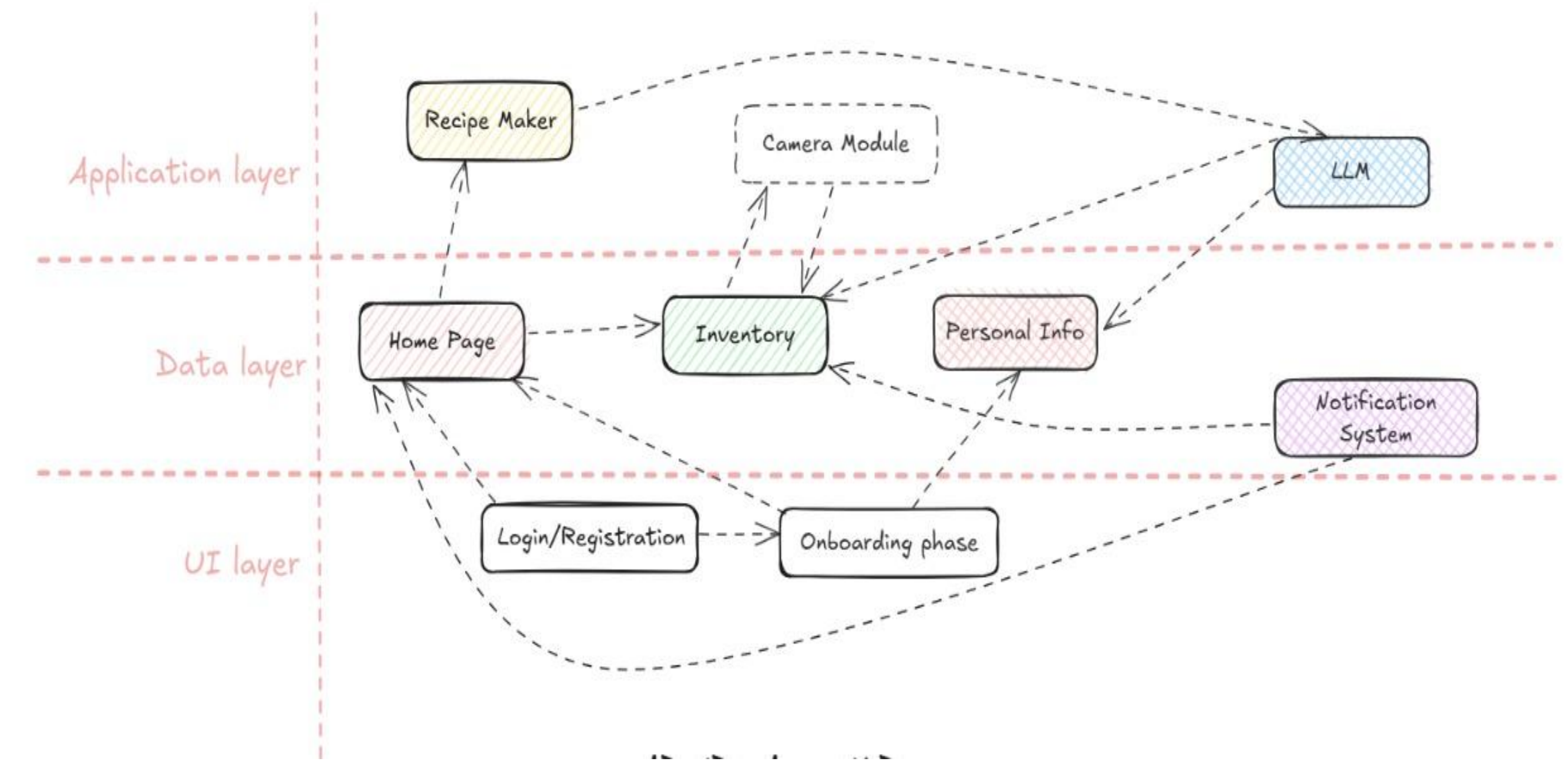
Login → Home → Inventory sync from local storage

Onboarding → Personal profile → Recipe engine personalization

Inventory updates → Notification triggers → User alerts

Recipe requests → LLM processing →

Formatted suggestions → Home display



Implementation Details and Test Plan

Technology Stack

Frontend: React Native with Material UI components

Storage: Firebase RTDB

Backend: Nextjs api

AI Integration: Gemini

Notifications: Native push notification services

Development Phases

Phase 1: Foundation & Core Authentication

- User authentication system with email/password and social login
- Secure session management and token refresh
- Basic app navigation structure and routing
- Local storage initialization (SQLite/AsyncStorage)
- Material UI component library integration
- Basic error handling and loading states

Phase 2: User Onboarding & Profile Management

- Multi-step onboarding interface with progress indicators
- Personal information collection (age, blood group, medical conditions)
- Allergy and dietary goal input forms with validation
- Profile data persistence and retrieval
- Profile editing capabilities for existing users
- Onboarding skip/resume functionality
- Integration tests for data flow between onboarding and storage
- MindMeal user testing for form clarity and completion rates

Phase 3: Inventory Management & Core Functionality

- Inventory database schema and CRUD operations
- Add/update/delete item interface with floating action buttons
- Item categorization and quantity tracking
- Expiration date monitoring and alerts
- Search and filter functionality for inventory items
- Bulk item management capabilities
- Inventory sync with backend services
- Performance testing for large inventory datasets
- User acceptance testing for inventory workflows

Phase 4: Recipe Engine & AI Integration (Weeks 11-14)

Chat-based recipe request interface
LLM API client implementation and error handling
Recipe suggestion algorithm with inventory matching
Recipe card display with ingredients, prep time, and nutrition info
Ingredient highlighting for available vs. needed items

Testing Strategy

Unit tests for each logic module
Integration tests for data flow between layers
ADHD user testing for interface clarity and cognitive load
Performance testing for offline functionality
Security audits for personal health data protection

Success Metrics

User engagement: Daily active usage and feature adoption rates
Effectiveness: Reduction in meal planning time and decision fatigue
Satisfaction: ADHD user feedback on interface simplicity
Reliability: Offline functionality and sync accuracy

Conclusions and Future Work

Lessons Learned

- Simplicity is paramount - every additional click or decision point creates friction
- Grouping notifications prevents alert fatigue while maintaining usefulness
- Local-first storage is essential for daily-use apps targeting ADHD users
- Conversational interfaces lower barriers to recipe discovery

Future Enhancements

- Visual barcode scanning to eliminate manual inventory entry
- Meal prep assistance with step-by-step cooking guidance
- Social features for sharing successful recipes with the ADHD community
- Integration with grocery delivery services for seamless shopping
- Expanded nutrition tracking and health insights

References

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